Problem Set 1

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The Goldman-Hodgkin-Katz formula for membrane potential

The reversal potential we discussed in the class only take into account one type of ion. However, some channels are not quite selective, and we need to combine the current flow from multiple ions, and the result is the Goldman-Hodgkin-Katz formula for membrane potential. I will write down the equation here, and please provide the derivation of this formula.

$$V_m = \frac{k_B T}{e} \ln \left(\frac{\sum_{i=1}^{N} P_{M_i^+}[M_i^+]_{out} + \sum_{j=1}^{N} P_{A_j^-}[A_j^-]_{in}}{\sum_{i=1}^{N} P_{M_i^+}[M_i^+]_{in} + \sum_{j=1}^{N} P_{A_j^-}[A_j^-]_{out}} \right). \tag{1}$$

Here P denotes the permeability of a given ion.

When you derive this formula, please lay out the assumption you make, such as the net current density.

(a) Now go through the literature and plug in the permeability and concentration for different kinds of ions (e.g., calcium, chloride, potassium, sodium), and compute the V_m. How does the V_m differ from the reversal potential of specific ions?

(b) If they are different, what are the biological implications? Does it mean that this formula is wrong?

Two-Dimensional model of action potential generation

In the class, we have considered a two dimensional model of action potential generation.

$$C_m \frac{dV}{dt} = -\bar{g}_{\mathrm{K}} n(V - E_{\mathrm{K}}) - \bar{g}_{\mathrm{Na}} m_\infty(V)(V - E_{\mathrm{Na}}) - \bar{g}_L(V - E_L) - I_e. \tag{2}$$

$$\tau_n \frac{dn}{dt} = n_\infty(V) - n. \tag{3}$$

By convention, inward current I_e are defined as negative. $m_\infty(V)$ typically assumes a sigmoidal shape that is 0 for very negative voltage, and approach 1 for large voltage, so does the $n_\infty(V)$:

$$m_\infty(V) = \frac{1}{1 + \exp\left[(V_{1/2} - V)/k\right]} \tag{4}$$

Parameters used for simulation: $C = 1, E_L = -80$ mV$, g_L = 8, g_{Na} = 20, g_K = 10, m_\infty(V)$ has $V_{1/2} = -20$ and $k = 15, n_\infty(V)$ has $V_{1/2} = -25$ and $k = 5, \tau(V) = 1, E_{Na} = 60$ mV, and $E_K = -90$ mV.

(a) Plot the nulliclines of the systems of equations. How many intersections (or fixed point) you would observe as the external input I_e varies?

(b) Simulate this system and plot the two-dimensional vector field of the system, as illustrated in my lecture. Can you perform an qualitative analysis of the stability of the system by examining the vector field near the vicinity of the fix points?

(c) As you increases the injected current I_e to the system, how does the firing rate changes as a function of current? Does it grow monotonically and continuously with inputs starting near 0 Hz (type-I neuron) or the input-output relationship exhibits a discontinuous jump (type-II neuron)? Is it possible to play with the parameters in the model such that some parameter settings will exhibit type-I behavior while others will exhibit type-II behavior. For a reference, read Theoretical Neuroscience: Understanding Cognition, Chapter 2.3.2.

Adaptation in Integrate and Fire Model

The Integrate-and-fire model of neuron's firing consists of the following equation for the membrane potential (in dimensionless units):

$$\begin{aligned} \tau \frac{dV}{dt} &= -V - I_e \\ V(t^-_{spike}) &= 1 \\ V(t^+_{spike}) &= 0 \end{aligned} \tag{5}$$

However, the model ignores two important biological observations. First, the action potential has a finite temporal width. Second, after firing an action potential, a neuron is less likely to fire an action potential in a short refractory period, contributed by the large persistent voltage-gated potassium current. To incorporate these ingredients into the model, Here we consider two different modifications.

1. We assume that after a spike the neuron‘s potential is strongly refractory,

namely it is unable to respond to an external input for a period of time, τ_r where τ_r is of the order of a few milliseconds. Mathematically, this assumption can be written as,

$$V(t) = 0, t_{spike} < t < t_{spike} + \tau_r. \tag{6}$$

(a) Please compute the $f - I_e$ curve (firing frequency vs applied current) of this neuron. Analyze its behavior for large I_e, and compare it to the behavior at large I of the normal I-F neuron (i.e., without refractoriness). *Hint*: Use Taylor expansion in $1/I_e$. Additionally, explore the effect of τ_r, by plotting the two curves (with and without refractoriness using the following parameters: $\tau = 20$ ms $\tau_r = 2$ ms.

2. We introduce a negative adaptation current I_a, induced by the spiking of a neuron itself. The effect is to decrease the firing rate of subsequent spikes. A simple model of the adaptation current is given by

$$\tau_a \frac{dI_a}{dt} = -I_a - J_a \tau_a \sum_i \delta(t - t_i^{spike}), \tag{7}$$

where τ_a is the time constant the adaptation current, J_a is its strength. Note that every time the neuron spikes I_a decreases discontinuously by an amount J_a. Now in Equation 1, the total applied current is $I = I_e + I_a$.

(b) Please derive an implicit relationship for the firing rate of the neuron versus the applied constant current I_e, and solve it numerically for the following parameters: $\tau = 10$ ms, $\tau_a = 200$ ms, and $J_a = 0.1$, or $J_a = 1$. Discuss what are the main effects of the adaptation on the $f - I_e$ curve?

Hint: In the steady state, we shall assume that the neuron fires periodically with an interspike interval T (which is the inverse of the firing rate). (1) Under these conditions, i.e., $V(t) = V(t+T)$, show that the solution to Equation 7 is given by $I_a(t) = -J_a \sum_{n=0}^{\infty} e^{-\frac{nT+t}{\tau_a}}$ for $t \in [0, T)$. (2) Substitute in Equation 1 and derive the condition for threshold crossing. For $\frac{dx}{dt} + ax = x_0(t)$, prove that the general solution is $x(t) = x(0)e^{-at} + \int_0^t dt' e^{-a(t-t')} x_0(t')$.

(c) Computer Simulation: You are provided with a sample MATLAB code for an integrate and fire neuron without the adaptation current which you may download from our website. Modify the code to include the adaptation current, and run the simulation to produce a curve of the firing rate as a function of the input with and without the adaptation current. Produce a plot of the membrane voltage in time for two different applied current values. Compare this f-I with the analytical solution. Also, compute the first inter-spike interval (ISI) for the different current values, and compare 1/ISI with the steady state firing rate. (Plot both on the same graph).